

Module 5 Claw Retrieval System: Team J-9

Michael W, Clint A, Jack D, Ruoyu Z, Sameen H,
Claus T

Design Challenge

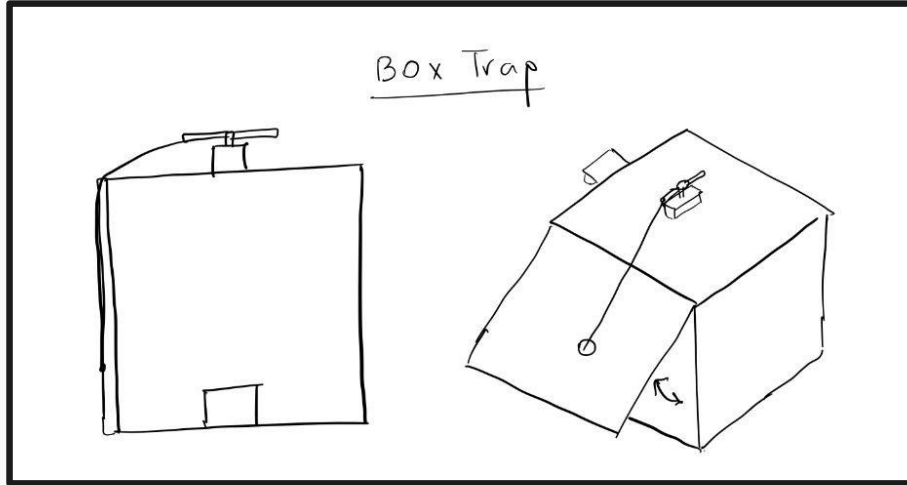
- Objective: Build a fully autonomous retrieval system to pick up, lift, and release a variety of objects
- Hardware Constraints: Limited to using one servo motor and one ultrasonic sensor
- Operational Constraint: Must be completely autonomous during operation

Overall Strategy

- Prototyped various gripping mechanisms (ex. Different arm counts, “teeth” grip, etc.)
- Considered risks related to these prototypes and mitigated them
- Evaluated designs based on simplicity, stability, and effectiveness
- Prioritized design diversity over early commitment to avoid being locked into a suboptimal solution

Initial Concepts and their Prototypes

"Box Trap" – Sameen H.



Needs

Picking up larger objects
securedly

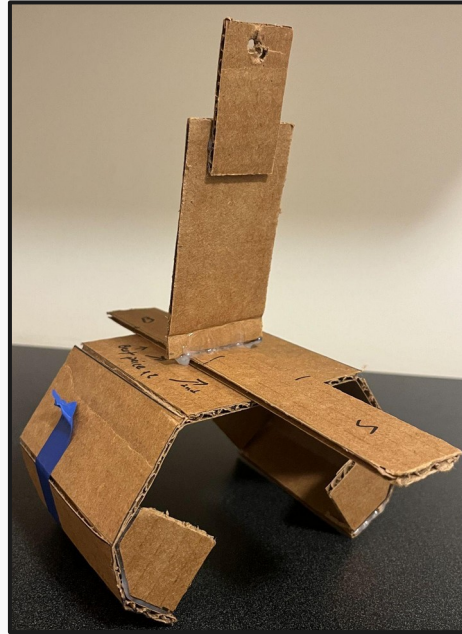
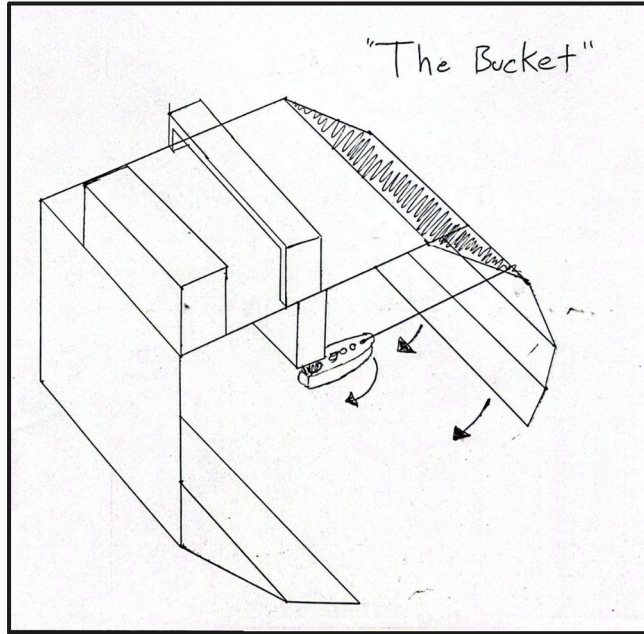
Features

Box-shaped enclosure
Single-flap opening

Insights

Difficult to use as object
does not fall out once
captured

"The Bucket" – Jack D.



Needs

Picking up longer objects

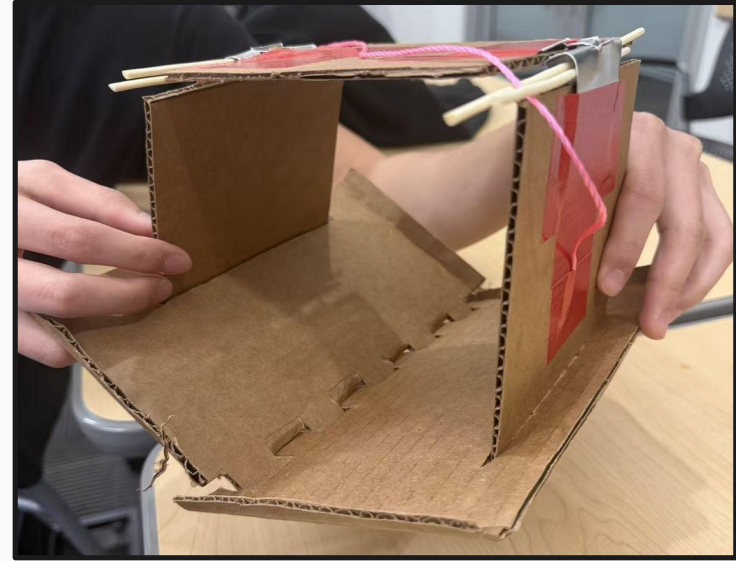
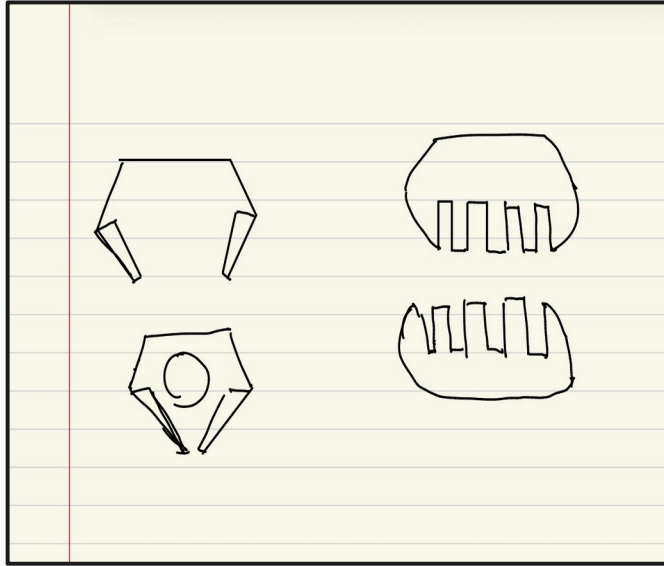
Features

Self-balancing and stabilizing arduino placement

Insights

Balancing component is essential to ensuring both arms close properly and in contact

"Walnut" – Michael W.



Needs

Picking up longer objects

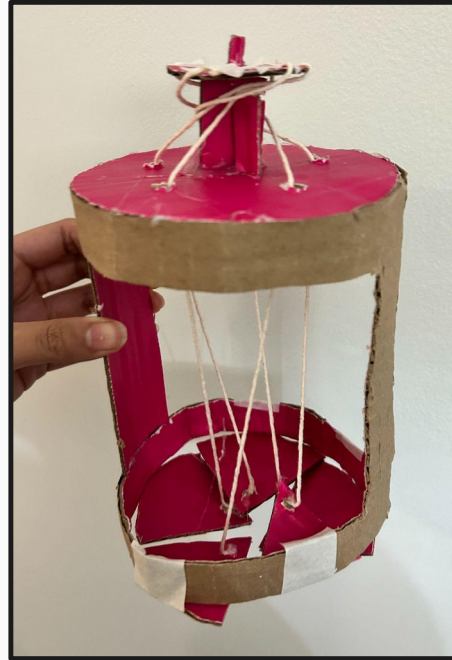
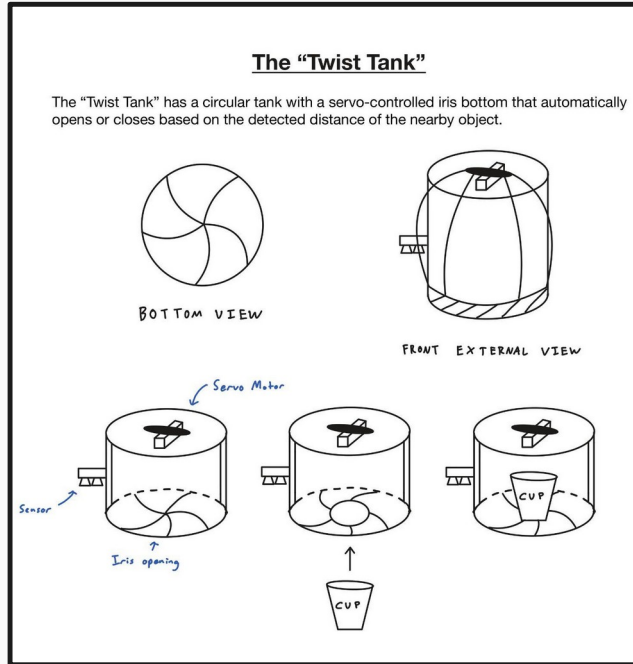
Features

Teeth-like claw design
Hinge joints

Insights

Teeth can help with grip, but
not significantly
Hinges create risk of claw
failure

The “Twist Tank” – Clint A.



Needs

Picking up smaller,
rounder objects

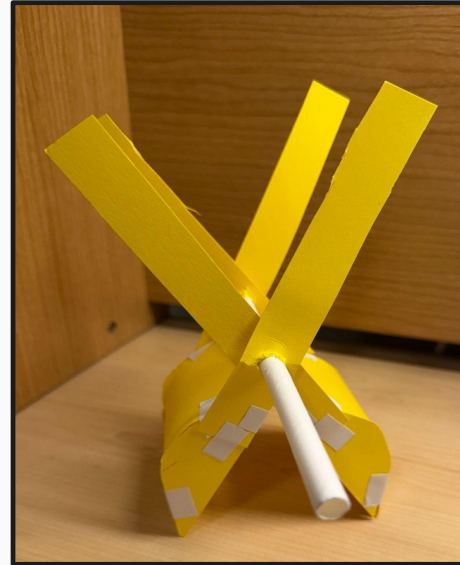
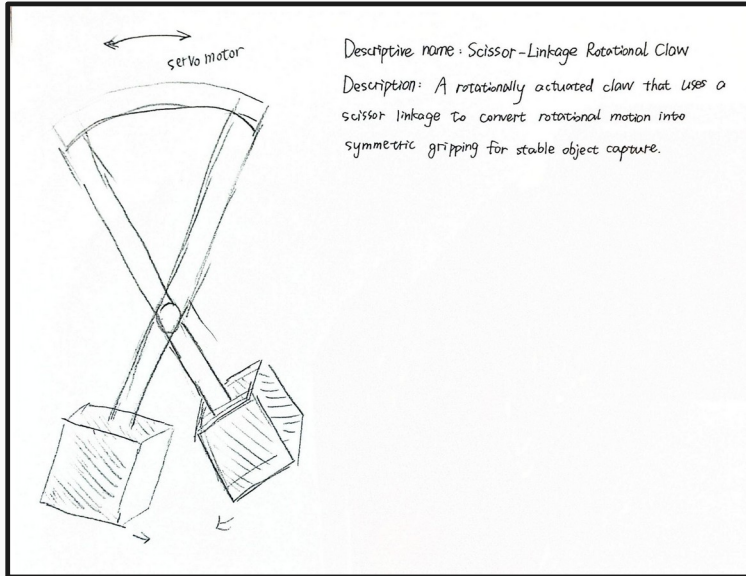
Features

Iris-like opening
Twisting grippers

Insights

Keeping strings close
together can become
knotted

"SLRC" – Ruoyu Z.



Needs

Picking up smaller, rounder objects

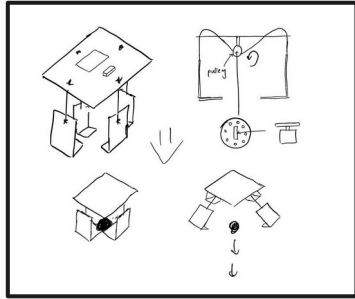
Features

Scissor-like joint connection
Symmetrical motion

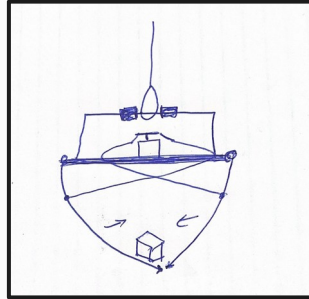
Insights

Helped us understand relationship between torque and arm length

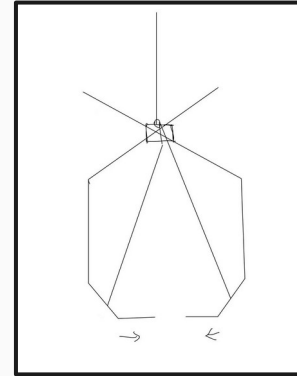
Alternative C-Sketches (not selected)



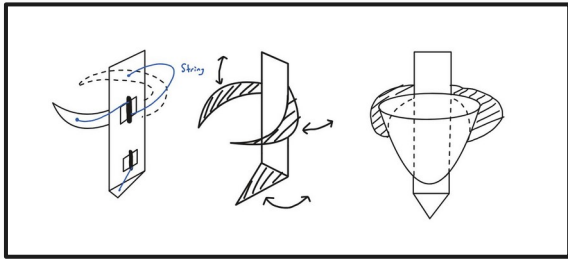
"Pulley Scooper" - Sameen H.



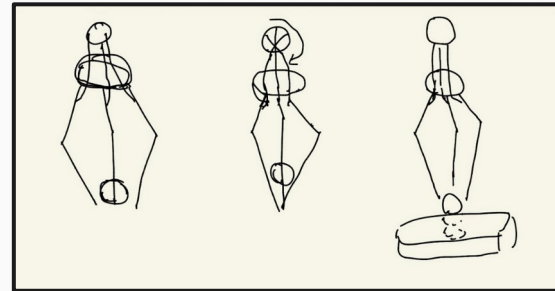
"String Scoop" - Jack D.



"Double-Walled Claw" -
Ruoyu Z.



"Tri-Claw" - Clint A.

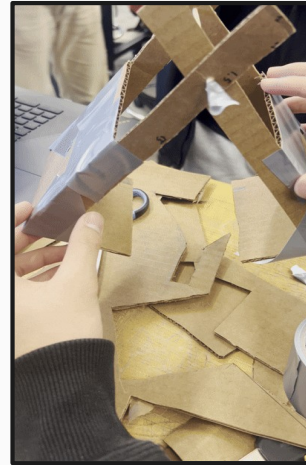
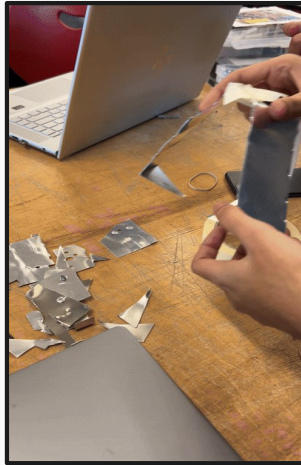
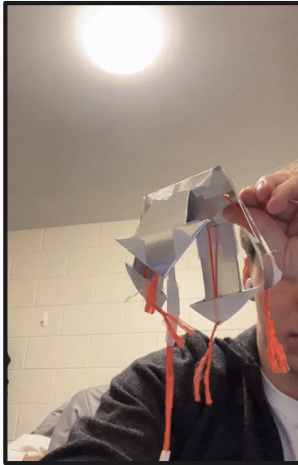


"Eagles" - Michael W.

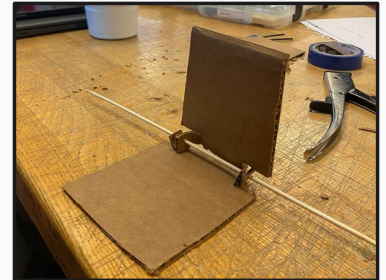
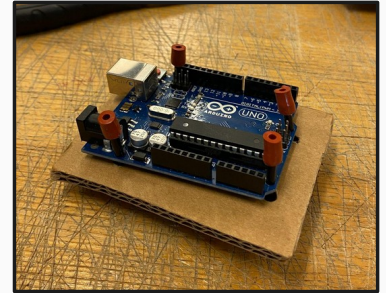
Focused/Comprehensive Prototypes

We created functioning prototypes to understand their ease of use

Claw Prototypes



Arduino Holder & Hinge



Risk Assessment and Management

Risk Classification Table

We initially identified our risks and classified them using a table

		Risk Source		
		External	Preventable	Strategic
Risk Category	Safety			
	Technical	Unstable object alignment during lifting	Fitment of the claw is not stable	Longer claws with insufficient torque
	Project Management		A member forgets to bring the sheetmetal to class	
	Operational	Metal sheet gets bent during competition	Wires come loose during the competition	

Risk Matrix

		Severity				
		1 (Insignificant)	2 (Minor)	3 (Moderate)	4 (Major)	5 (Severe)
Likelihood	5 (Almost Certain)					
	4 (Likely)			[12] Create longer flaps which reduces torque while lifting heavier objects		
	3 (Possible)		[10] Member(s) miss the final competition		[6] Unstable object alignment during lifting	[11] A member forgets to bring the sheetmetal
	2 (Unlikely)			[2] Servo motor doesn't work to full capacity [5] Fitment of claw is not stable	[7] Metal sheet gets bent during competition	[8] Not getting project finished by last studio [9] Not getting project poster done by due date
	1 (Rare)			[3] Sensor doesn't correctly gauge readings	[4] Battery is not charged	[1] Code fails to compile

Team Actions to Manage Major Risks

Risk	Mitigation	Score
Loose wiring during competition	Tape & secure; verify in testing	12
Metal bending during competition	Inspect material; pre-test strength	8
Project not finished on time	Weekly plan & checklist tracking	10
Poster not completed by deadline	Assign tasks; weekly check-ins	10
Missing sheet metal in class	Phone reminder before studio	15
Limited metal supply	Reuse metal for secondary systems	12

Prototype Investigations

We conducted investigation on our most significant technical risks

→ Prototype Investigation 1

Investigate the technical risk of insufficient torque

→ Prototype Investigation 2

Focusing on the risk of unstable object alignment during lifting.

Investigation 1 - Gripping force

Procedure

Lift objects of increasing mass

We have claws designed with teeth or tape

Records success

Result

Tape design (5) vs Teeth design (3)

The teeth design does not improve the grip

Investigation 2 - Object Alignment

Procedure

Lift objects in different orientation

We have three claws with different width

Record success or failure

Result

15.24cm width(4) vs 14.20cm(3) vs 13.03cm(3)

Widest claw is more versatile

Final Claw Decision and Justification

Weighted Decision Matrix

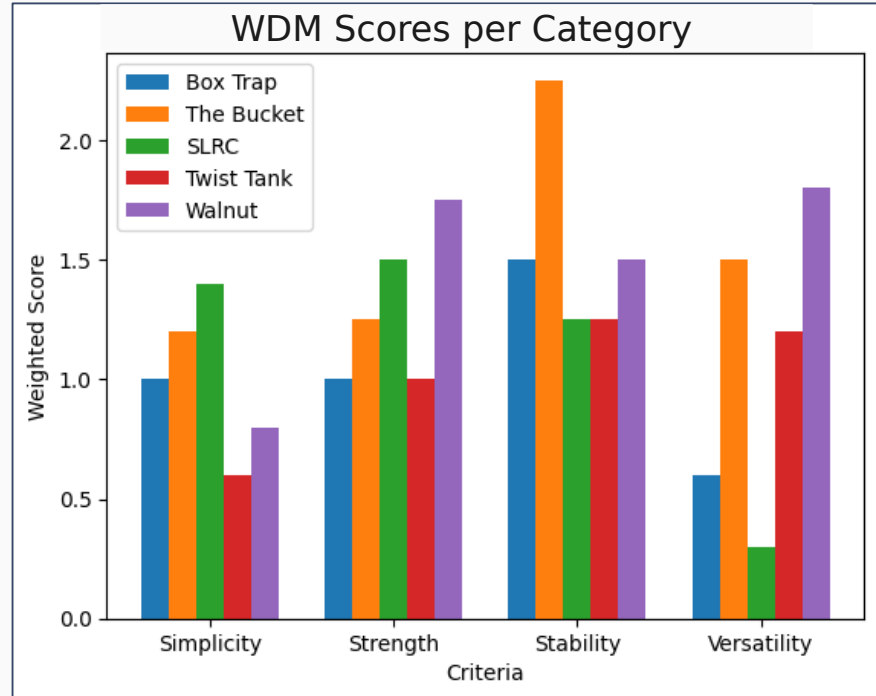
Weights:

Simplicity
(20%)

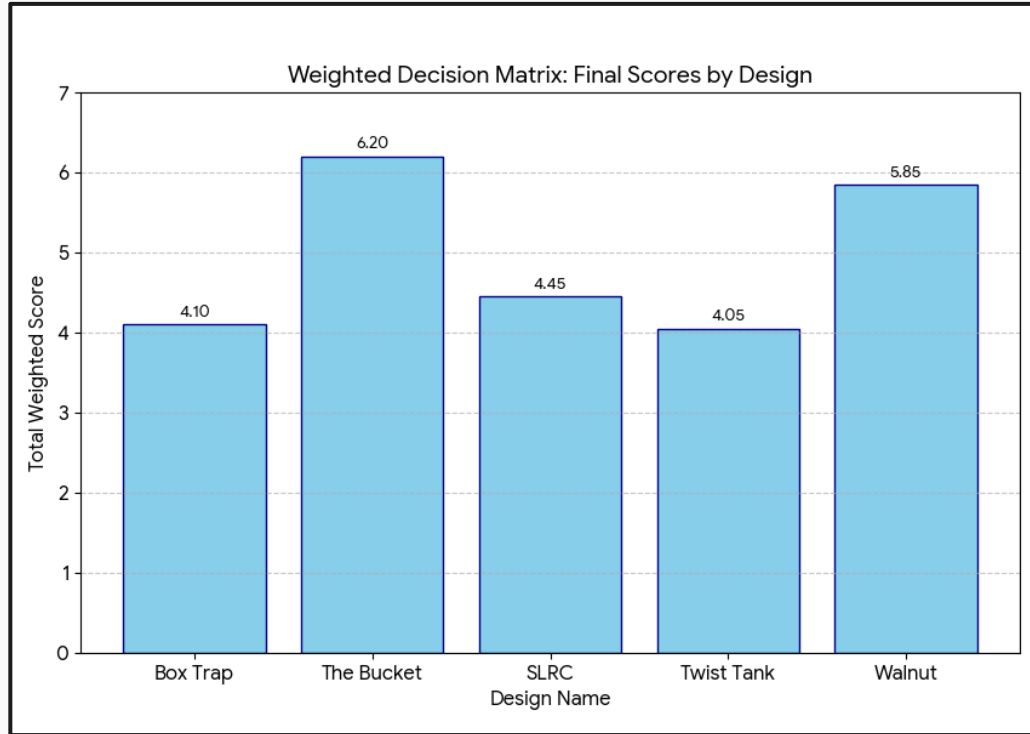
Strength
(25%)

Stability (25%)

Versatility
(30%)



Weighted Decision Matrix

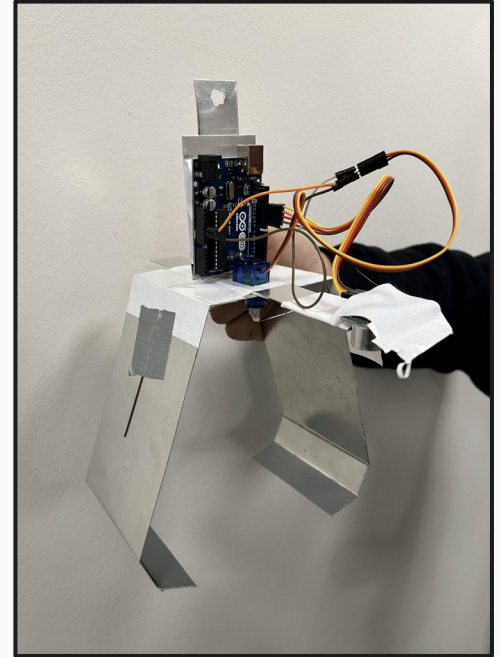


Final Score: The Bucket > Walnut > SLRC > Box Trap > Twist Tank

Design Decisions

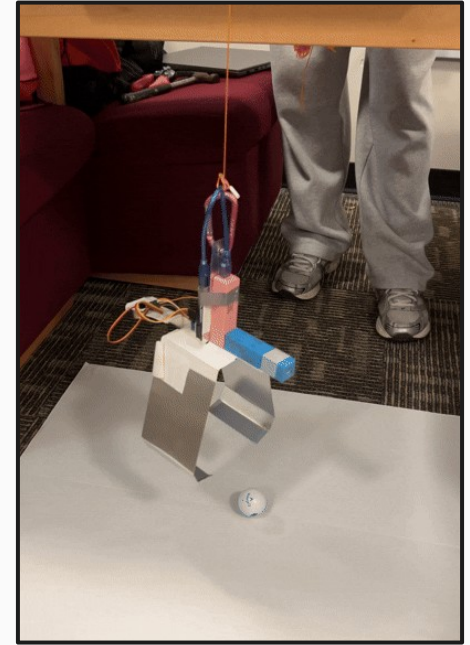
Selected Final Design: **The Bucket**

1. Highest adaptability and picks up any object (except heavy ones)
2. Enclosure geometry reduces slippage of smaller objects and improves stability
3. Simple design, with two claws to pick up long and big objects

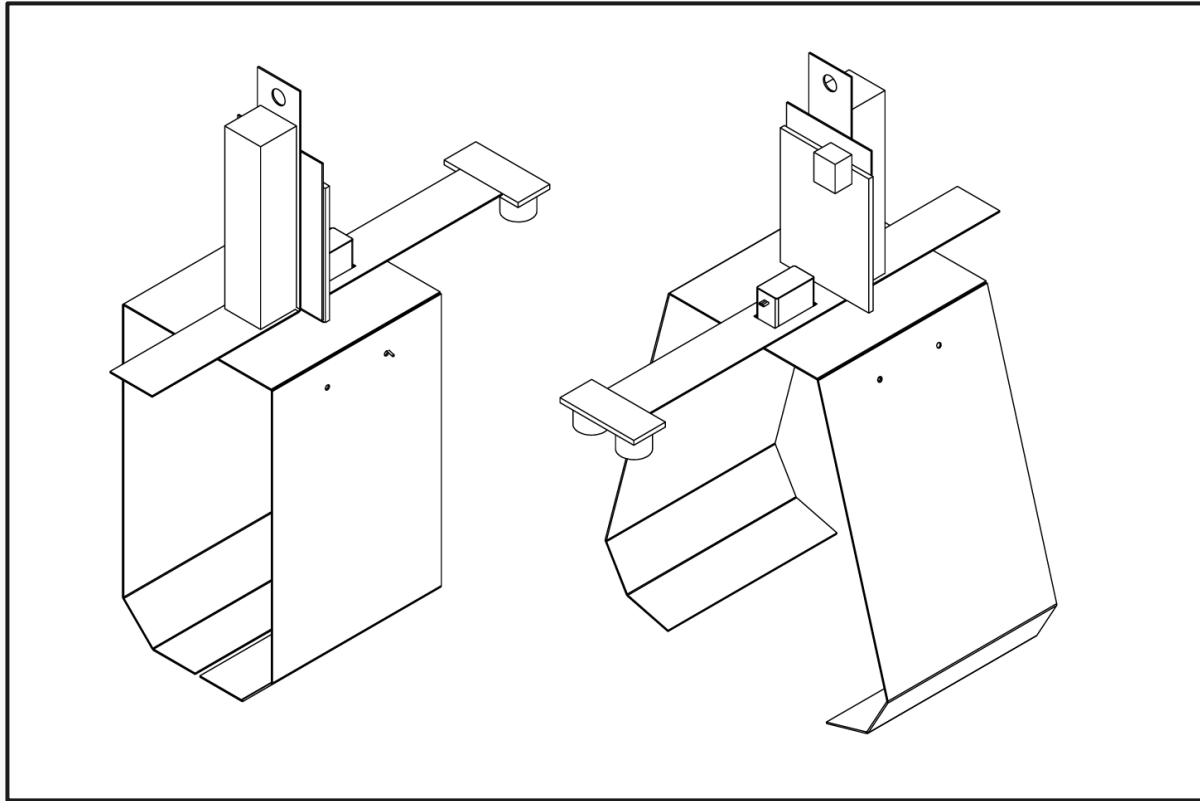


Features of “The Bucket”

- Servo-driven dual flaps (60° open/close)
- Bent flap edges prevent object escape
- Lightweight structure → easy balance
- Centralized arduino & battery for stability



Final Design (Isometric Views)



Competition Performance & Reflection

Round One: 7 objects picked up

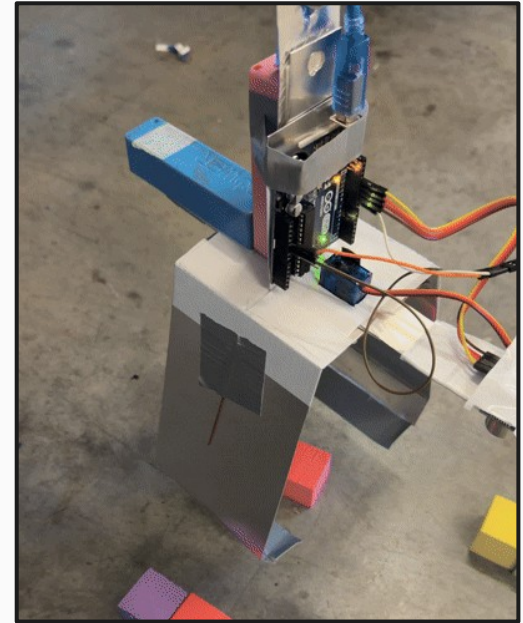
- Longer handle produced little torque which was a strategic risk that should have been more strongly considered

Round Two: 16 blocks picked up

- Exceeded expectations through great teamwork
- Lightweight and stable structure allowed objects to be moved quickly

Round Three: All but one block put in the middle

- Overall high efficiency, moved larger lighter objects



Thanks!

Team J-9: Michael W, Clint A, Jack D,
Ruoyu Z, Sameen H, Claus T

Use of GenAI

OpenAI. (2025). *ChatGPT* (GPT-5)
[Large language model].

<https://chatgpt.com/>

Our uses of GenAI:

- Improve grammar and check spelling
 - Improve slide formatting

Contributions

Ruoyu	Slides 1-3
Clint	Slides 4-9
Michael	Slides 10-12, 16-17
Sameen	Slides 12-15
Jack	Slides 18-20